Lecture 5
Data Warehouse and OLAP Technology

CSC1104: Data Warehousing and Data Mining
2026/2027 Semester 1

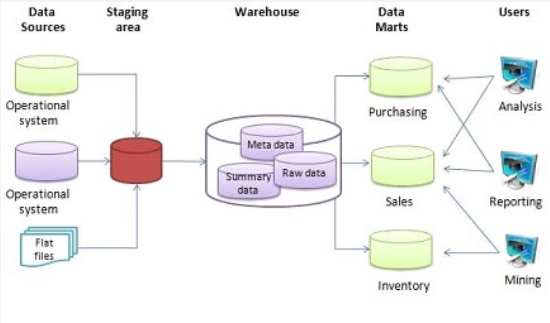
Prof. Mark Roantree
COMSCI4 @ Dublin City University

Basic Concepts
Data Cubes and OLAP
Warehouse Design
Step 1. Select the Process
Step 2. Declare the Grain
Step 3. Choose the Dimensions
Step 4. Identify the Facts.
Warehouse Implementation
Integration and Interoperability
Conformed Dimensions
Conformed Facts

Agenda

1. Basic Concepts
2. Data Cubes and OLAP
3. Warehouse Design
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Basic Concepts



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What is a Data Warehouse?

- A decision support database that is maintained separately from the organisation's operational database.
- Support information processing by providing a solid platform of consolidated, historical data for analysis.
- Data Warehousing: The process of constructing and using data warehouses.

Inmon's Definition

A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process.

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Data Warehouse: Subject-Oriented

- Organized around major subjects, such as customer, product, sales
- Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing
- Provide a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process

Inmon's Definition

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Data Warehouse: Integrated

- Constructed by integrating multiple, heterogeneous data sources
  relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
  Ensure consistency in naming conventions, encoding structures, attribute measures, among different data sources
  Hotel price: currency, tax, breakfast covered, etc.
  When data is moved to the warehouse, it is converted.

Inmon's Definition

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Data Warehouse: Time Variant

- The time horizon for the data warehouse is significantly longer than that of operational systems
  - Operational database: current value data
  - Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
  - Contains an element of time, explicitly or implicitly
  - But the key of operational data may or may not contain *time element*

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Data Warehouse: Non-volatile

- A physically separate store of data transformed from the operational environment
- Operational update of data does not occur in the data warehouse environment
  - Does not require transaction processing, recovery, and concurrency control mechanisms
  - Requires only two operations in data accessing: initial loading of data and access of data

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OLTP vs. OLAP

Feature	OLTP	OLAP
Characteristic	operational processing	informational processing
Orientation	transaction	analysis
User	clerk, DBA, database professional	knowledge worker (e.g., manager, executive, analyst)
Function	day-to-day operations	long-term informational requirements, decision support
DB design	ER based, application-oriented	star/snowflake, subject-oriented
Data	current; guaranteed up-to-date	historical; accuracy maintained over time
Summarization	primitive, highly detailed	summarized, consolidated
View	detailed, flat relational	summarized, multidimensional
Unit of work	short, simple transaction	complex query
Access	read/write	mostly read
Focus	data in	information out
Operations	index/hash on primary key	lots of scans
Number of records accessed	tens	millions
Number of users	thousands	hundreds
DB size	100 MB to GB	100 GB to TB
Priority	high performance, high availability	high flexibility, end-user autonomy
Metric	transaction throughput	query throughput, response time

Figure 1: OLTP vs. OLAP: A Comparison



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Properties of a Warehouse: Data Engineering

- Integrated data
- Detailed and summarised data
- Historical data
- Metadata



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Integrated Data

- **Goal**: Integrating multiple information systems aims to combine them so they provide the illusion of interacting with a single information system.
- **Benefit**: Users are provided with a **homogeneous logical view** of data that is physically distributed over heterogeneous data sources.
- **Task**: Detection and resolution of schema and data conflicts regarding structure and semantics.
- Without integrated data, too much time is spent cleaning and transforming data before data mining can commence effectively.
- Keys must be reconstituted;
  encoded values reconciled;
  data structures standardized;
  and many more tasks before data mining.



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Detailed and Summarised Data

- **Goal**: Store both detailed and summarised data: (heavily) duplicate data.
- **Benefit** 1: Detailed data is necessary when the data scientist wishes to examine data in its most *granular* form.
- Very low levels of detail hide important patterns that require the study of very low level (detailed) data.
- **Benefit** 2: No need to re-compute expensive queries.
- Summarised data ensures the reuse of a prior analysis (using a pre-existing *cube*).
- Data scientist can reuse the work of others rather than build everything from scratch.
- **Task**: Store prior queries, especially COUNT, SUM, GROUP-BY which represent **dimensional** queries.



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Historical Data: Longer Term Forecasting

- **Goal**: Make the Data Warehouse a Read-Only database: Delete Nothing!
- **Benefit**: Historical data is important to the data scientist for training machine learning models.
- A data scientist working only with very current information cannot detect trends and long-term patterns of behavior.
- Historical information is crucial to understanding the seasonality of business or other machine learning tasks.
- **Time** is an important dimension.



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Metadata

- **Goal**: Metadata serves as a *road map* to the data scientist, who uses metadata to describe the *context* of information.
- Metadata is not simply structure (intensional data in relational database theory).
- **Benefit**: When examining information *over time*, context becomes as relevant as content.
- Raw data becomes very difficult to work with when there is no explanation for the *meaning* of the data.
- It is often very difficult to interpret the results of machine learning experiments without context.



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Why a Separate Data Warehouse?

- High performance for both systems
  - DBMS tuned for OLTP: access methods, indexing, concurrency control, recovery
  - Warehouse tuned for OLAP: complex OLAP queries, multidimensional view, consolidation
- Different functions and different data
  - missing data: Decision support requires historical data which operational DBs do not typically maintain
  - data consolidation: DS requires consolidation (aggregation, summarisation) of data from heterogeneous sources
  - data quality: different sources typically use inconsistent data representations, codes and formats which have to be reconciled



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Data Warehouse: A Multi-Tiered Architecture

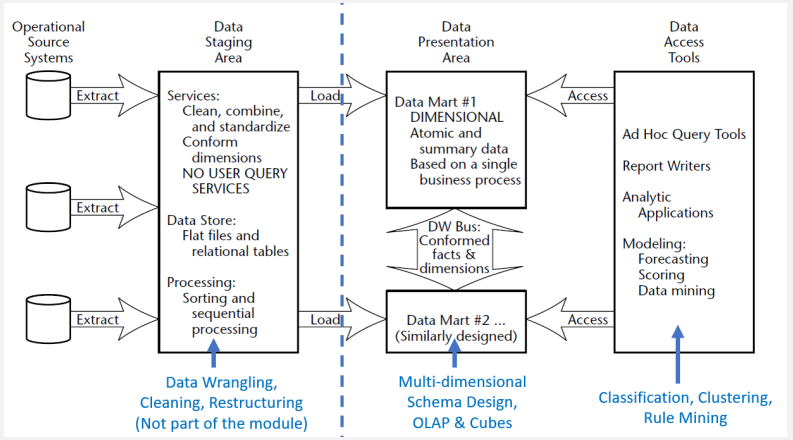


Figure 2: ETL Architecture



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Extraction, Transformation, and Loading (ETL)

- **Data extraction**: get data from multiple, heterogeneous, and external sources
- **Data cleaning**: detect errors in the data and rectify them when possible
- **Data transformation**: convert data from legacy or host format to warehouse format
- **Load**: sort, summarize, consolidate, compute views, check integrity, and build indicies and partitions
- **Refresh**: propagate the updates from the data sources to the warehouse



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Metadata Repository

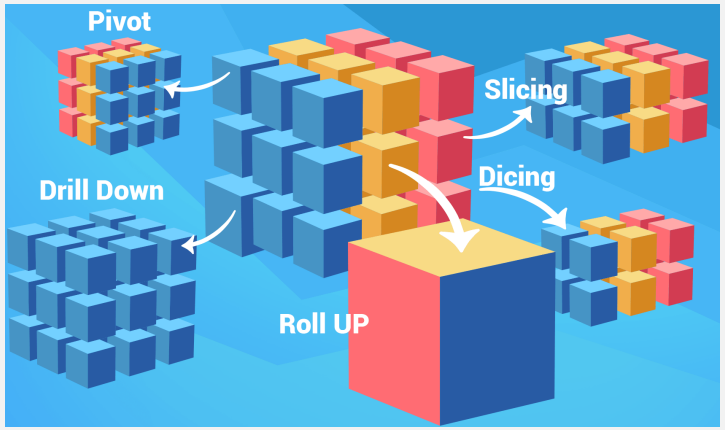
What are the different types of metadata?

- Descriptions of: data warehouse schema (eg. Slide 5.25),
  dimensions (Slide 5.25),
  hierarchies (Slide 5.32),
  data mart contents (query description).
- Operational metadata: data lineage (history of migrated data and transformation path eg. comes from Payroll database and is transformed to Net Income),
  currency of data (active, archived, or purged),
  monitoring information (warehouse usage statistics, error reports, audit trails eg. every change made to a stored query)
- Summarisation Algorithms (that create stored queries).
- Mappings from operational databases to the data warehouse (can be part of operational metadata).
- System performance: precomputed queries and their



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Data Cubes and OLAP





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From Tables and Spreadsheets to Data Cubes

- A data warehouse is based on a **multidimensional** data model which views data in the form of a **data cube**.
- A data cube, such as sales, allows data to be modeled and viewed in multiple dimensions:
  - **Dimension tables**, such as item (item_name, brand, type), or time (day, week, month, quarter, year);
  - **Fact tables**, containing measures (such as amount_sold) and keys to each related dimension tables.
- In data warehousing, an *n*-D base cube is called a **base cuboid**. The top most *0*-D cuboid, which holds the highest-level of summarisation, is called the **apex cuboid**.
- The *lattice* of cuboids forms a data cube.



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Cube: A Lattice of Cuboids

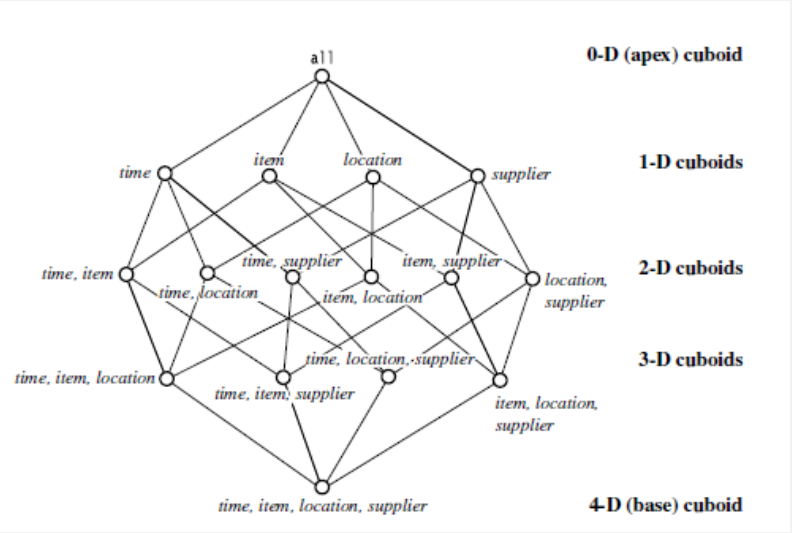


Figure 3: 4-D Cube and its Cuboid Parts

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Understanding the Lattice and Dimensionality

A 4-dimensional lattice has 5 levels of data computations.

- Apex contains a single value.
- 1-D contains 4 values, summarisations across each of the 4 dimensions.
- 2-D contains 6 values, summarisations across all possible pairwise combinations.
- 3-D contains 4 values, summarisations across all possible triple combinations.
- 4-dimensional Base contains a single value.
- However, this hides the true dimensionality as it assumes only a single value for each cuboid.

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Conceptual Modeling of Data Warehouses

Modeling data warehouses: **dimensions** & **measures**

- **Star schema**: A *fact* table in the middle *connected* to a set of dimension tables.
- **Snowflake schema**: A star schema refinement where a dimensional hierarchy is *normalised* into a set of smaller dimension tables, forming a shape similar to a snowflake.
- **Fact constellations**: Multiple fact tables *share dimensions*, viewed as a collection of stars, called a *galaxy* schema or *fact constellation*.

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Star Schema

- The most common modeling paradigm is the **star schema**, in which the data warehouse contains
  1. a large central table (fact table) containing the bulk of the data, with no redundancy, and
  2. a set of smaller attendant tables (dimension tables), one for each dimension.
- The schema graph resembles a starburst, with the dimension tables displayed in a radial pattern around the central fact table.

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Star Schema for a Sales Data Warehouse

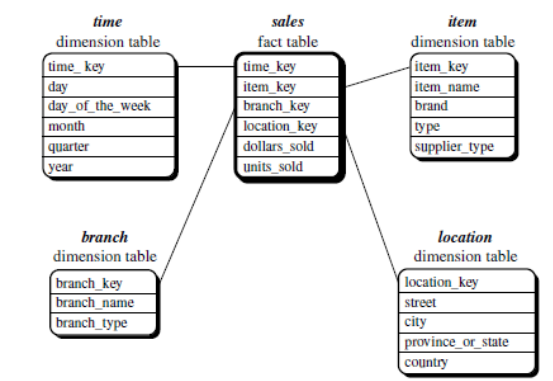


Figure 4: AllElectronics Star Schema

- Sales are considered along four dimensions: *time*, *item*, *branch*, and *location*.
- The schema contains a central **fact table** for *sales* that contains keys to each of the four dimensions.
- Contains two measures: *dollars sold* and *units sold*.

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Star Schema Model

- Each **dimension** is represented by a single table of attributes.
- Every dimension is connected to the fact table through *PK-FK* relationships.
- The *location* dimension table contains the attribute set *{location key, street, city, province or state, country}*.
- Redundancy (non 3-NF) is normal: {Dublin 1, Dublin, Ireland} can be repeated many times so values for *City* and *Country* can be repeated many times.
- Attributes within a dimension may form either a hierarchy (total ordering) or a lattice (partial ordering).

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Snowflake Schema for a Sales Data Warehouse

The **snowflake schema** is a variant of the star schema model, where some dimension tables are normalised splitting dimensional data into additional tables.

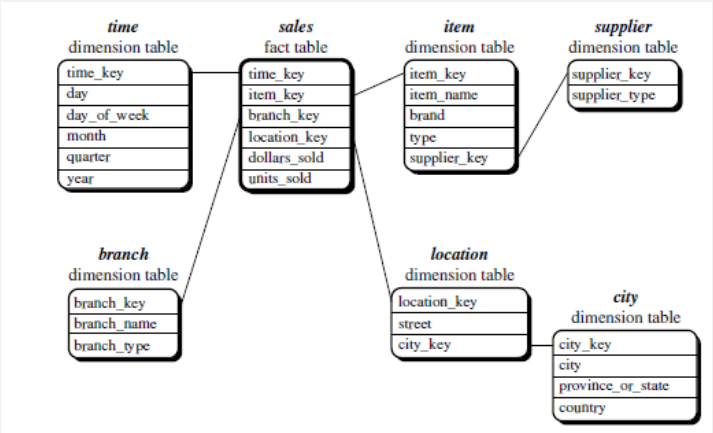


Figure 5: Snowflake Schema for *Sales* connected to 4 dimensions using their PKs stored in the fact table as FKs.

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Snowflake Schema

- The major difference between the *snowflake* and *star* schema models is that snowflake dimensions may use normalisation to reduce redundancies.
- However, this saving of space is negligible in comparison to the typical magnitude of the fact table.
- Furthermore, the *snowflake* structure can reduce the performance, since more joins will be needed to execute a query.
- Although the *snowflake* schema reduces redundancy, it is not as popular as the *star* schema in data warehouse design.

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Fact Constellation

- Sophisticated applications may require multiple fact tables to share dimension tables.
- This kind of schema can be viewed as a collection of stars, and hence is called a galaxy schema or a fact constellation.
- The schema in figure 6 specifies two fact tables: *sales* and *shipping*.

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Fact Constellation Schema for Sales and Shipping

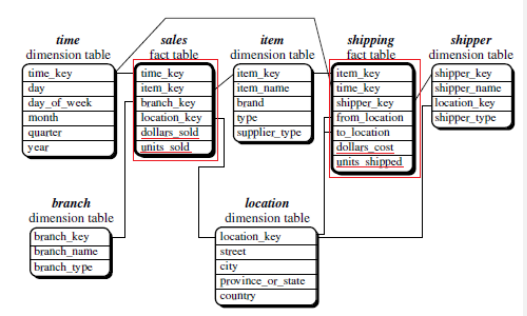


Figure 6: Sample Fact Constellation Schema

- Shipping has 5 dimensions (keys): *item key, time key, shipper key, from location*, and *to location*, and two measures: *dollars cost* and *units shipped*.
- A *fact constellation* schema allows dimension tables to be shared between fact tables.
- For example, dimensions for *time, item,* and *location* are shared by both the *sales* and *shipping* facts.

Concept Hierarchy

- A **concept hierarchy** defines a sequence of mappings from a set of low-level concepts to higher-level, more general concepts.
- Consider a concept hierarchy for the dimension location: City values for location include *Vancouver*, *Toronto*, *New York*, and *Chicago*.
- Each city can be mapped to its province or state.
- Provinces and states can in turn be mapped to their country.
- These mappings form a concept hierarchy for the dimension location, mapping a set of low-level concepts (cities) to higher-level, more abstract concepts (countries).

Location Concept Hierarchy

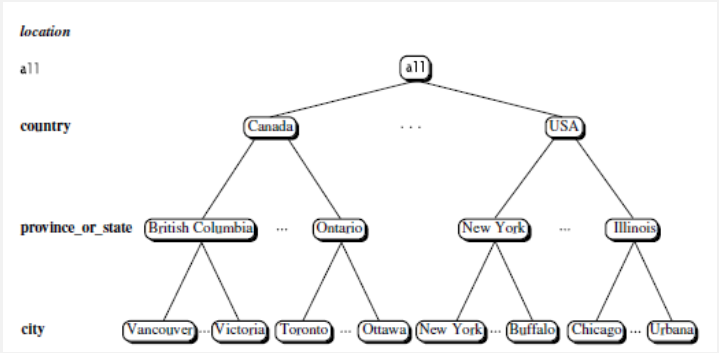
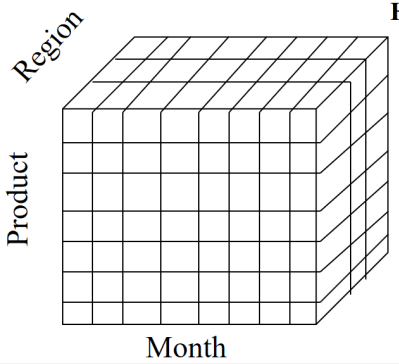


Figure 7: Concept Hierarchy for Location Dimension

Multidimensional Data

Sales volume as a function of product, month, and region



Dimensions: ***Product, Location, Time***
Hierarchical summarization paths

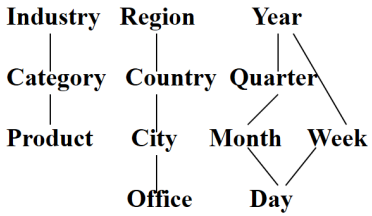


Figure 8: Sales by Product, Time, Geo-Location

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Cube: Multidimensional Construct

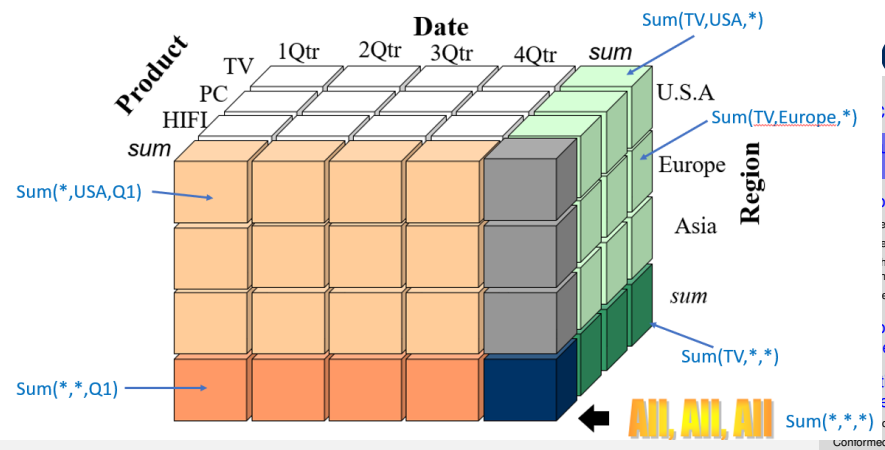


Figure 9: Cube Rollups

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OLAP Operations

- **Roll up**: summarise data by climbing up the hierarchy or by dimension reduction.
- **Drill down**: reverse of roll-up from higher level summary to more detail, or introducing new dimensions.
- **Slice and dice**: project and select.
- **Pivot** (rotate): reorient the cube, visualisation, 3D to series of 2D planes.
- **Drill across**: involving (across) more than one fact table.
- **Drill through**: through the bottom level of the cube to its back-end relational tables (using SQL).

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Rollup Operation

School	Module	Year	1st			
Computing	CA119	2018	27			
Computing	CA218	2018	55			
Computing	CA336	2018	10			
Computing	CA119	2019	7			
Computing	CA218	2019	21			
Computing	CA336	2019	27			
BioTech	BT160	2018	20			
BioTech	BT161	2018	30			
BioTech	BT162	2018	40			
BioTech	BT160	2019	45			
BioTech	BT161	2019	2			
BioTech	BT162	2019	12			
Business	CA119	2018	5			
Business	CA218	2018	10			
Business	BT162	2018	6			
Business	CA119	2019	10			
Business	CA218	2019	6			
Business	BT162	2019	6			
Rollup (Date)						
Computing	CA119	*	All Computing/CA119 are summed			
Computing	CA218	*	All Computing/CA218 are summed			
Computing	CA336	*				
BioTech	BT160	*				
BioTech	BT161	*				
BioTech	BT162	*				
Business	CA119	*				
Business	CA218	*				
Business	BT162	*				
Rollup(ALL)						
*	*	*	All cubes are summed			

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Typical OLAP Operations

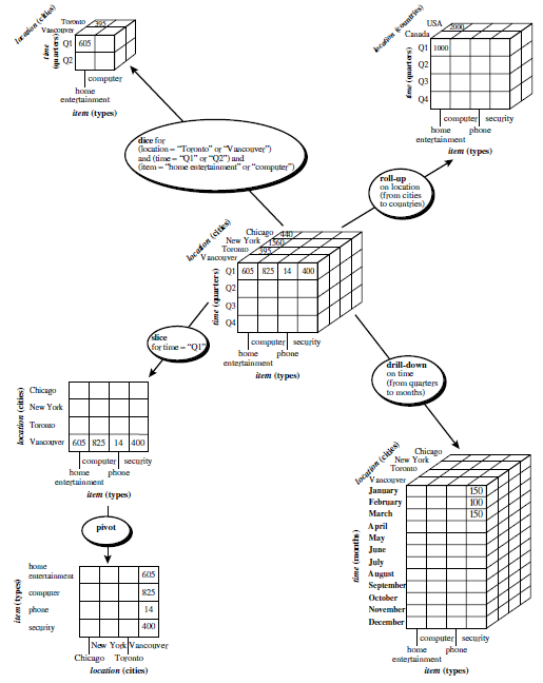
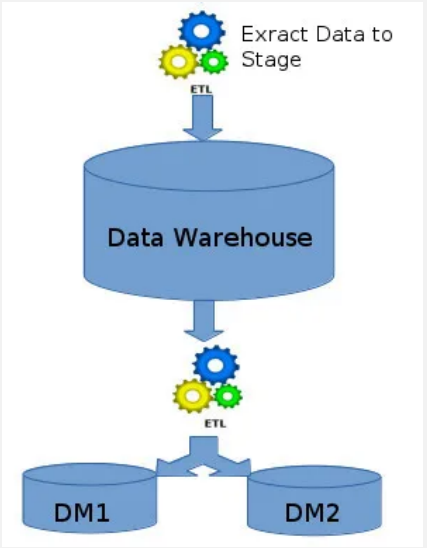


Figure 10: OLAP Operations

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Warehouse Design



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Four-Step Dimensional Design Process

1. **Select the Process**: What process do you wish to measure?
2. Declare the grain (*daily v weekly v monthly* or *Eircode v City v Province*) of the business process
3. Identify dimensions that apply to every row in the Fact Table.
4. Identify the numeric facts that populate the Fact Table.

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Step 1. Identify/Select the Process

- **Business measurement** processes provide the performance measurements users need to analyse in the data warehouse.
- What do you want to predict (for next week/month/year) that you need the data for?
- Example business processes:
  How many Female students taking csc1204, who live in Munster, are scoring firsts?
  How many frozen products, imported from France, are sold from January to April?
  How many patients, in Munster, tested positive for flu in December 2025? (these are dimensional queries!)

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Consistent Information

- Predictive algorithms require appropriate and correct data.
- Build a *single* dimensional model for every process in the organisation: all departments use the same model (schema).
- Focus on processes/activities (not departments) to deliver *consistent* information more efficiently throughout the organisation.
- Focus on students (not finance or student affairs) to deliver exam marks by module, by school, by faculty . . .
- *Consistent*: this important property is studied in Fact & Dimension Conformance later in this section.



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Step 2. Declare the Grain (appropriate level of detail)

- Specify *exactly* what an individual fact table row represents.
- The **grain** conveys the level of detail associated with the fact table measurements.
- It provides the answer to the question: *How do you describe a single row in the fact table?*
  For our 3 examples, **grain is**:
  - Every mark for every module for every student
  - Every item sold (units), per country of origin, by month, by sales location
  - Every positive test, for each disease, for a particular year in a specified region.



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Sample grain declarations

- An individual line item on a supermarket receipt as captured by a scan
- A line item on a bill received from a builder
- An individual boarding pass to get on a flight
- A daily snapshot of the stock levels for each product in a warehouse
- A monthly snapshot for every bank account



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Step 3. Choose the Dimensions (for every measure)

- Dimensions emerge from asking:
  1. How do analysts describe the data that results from the business process?
  2. At what level do scientists discuss the output from test sites?
  3. How are decisions made in hospitals when allocating . . . ?
- Once the grain is known, dimensions can be identified easily.
- **Enrich** fact tables with a robust set of dimensions capturing all variables needed for analysis. Dimensions hold the capability for powerful analyses.
- For each dimension: list all discrete, text-like attributes to populate each dimension table (Ireland, Yellow, Pass . . .)
- *What dimensions were used in our 3 examples?*



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Step 4. Identify the numeric facts that will populate each fact table row.

- Facts are determined by answering the question: *What are we measuring?*
- All candidate facts in a design must be true to the grain defined in step 2.
- Facts that clearly belong to a different grain *must be in a separate fact table*.
- Typical facts are numeric additive figures such as items sold, tests conducted, students who participated, amount spent . . .

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Summary

Consider both business users' requirements *and* realities of source data, to make decisions regarding the 4 steps.

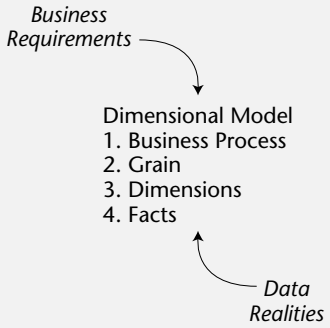
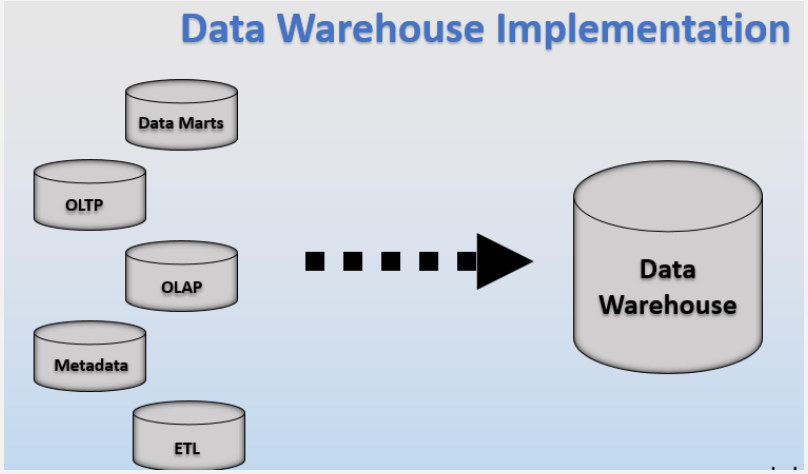


Figure 11: Key input to the four-step dimensional design process

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Data Cubes and OLAP



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Warehouse Implementation

- Which implementation is best?
- The company has 15 systems using database storage. How do I know which data source to include first?
- How do I know which data source to include next?
- Can I adopt an enterprise-wide plan (every uses the same schema)?

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Data Warehouse Bus Architecture

- By defining a *standard bus interface* for the data warehouse environment, separate data marts can be implemented by *different* groups at *different* times.
- The separate data marts can be integrated if they adhere to the standard (every uses the same schema).
- This approach adopts the *data warehouse bus architecture*.



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Design Methodology

The bus architecture provides a rational approach to decomposing enterprise warehouse planning.

1. Design a **master suite** of standardised dimensions and facts with **uniform interpretation** across the enterprise.
2. This establishes the *data architecture framework*.
3. Build the first (next) data mart that closely adheres to the architecture.
4. Repeat Step 3 until enough data marts exist to deliver an *integrated* enterprise data warehouse.



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Benefits: Reduce the Scale of the Problem

- The bus architecture offers the best of both worlds.
- They have an architectural framework that guides the overall design, but the problem is divided into manageable data mart implementation efforts.
- Separate data mart development teams follow the architecture guidelines while working fairly independently and asynchronously.
- If designed coherently, marts share a **uniform architecture** of *conformed* dimensions and facts for seamless integration.



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Data Warehouse Bus Matrix

- The rows of the bus matrix correspond to data marts.
- Create rows where sources/processes are different or where a process is too big.

	COMMON DIMENSIONS							
BUSINESS PROCESSES	Date	Product	Store	Promotion	Warehouse	Vendor	Contract	Shipper
Retail Sales	X	X	X	X				
Retail Inventory	X	X	X					
Retail Deliveries	X	X	X					
Warehouse Inventory	X	X			X	X		
Warehouse Deliveries	X	X			X	X		
Purchase Orders	X	X			X	X	X	X

Figure 12: Sample Data Warehouse Bus Matrix



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Constructing the Bus Matrix

- The tool used to *create*, *document*, and *communicate* the bus architecture is the **data warehouse bus matrix**.
- Working in tabular fashion, lay out the business processes of the organisation as matrix rows (should be sources of data).
- Matrix rows translate into **data marts** based on the organisation's primary activities.
- Begin with data marts derived from a single primary source, known as *first-level data marts*.



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First Level Data Marts

- *Step 1*. Begin with *first-level data marts* as they minimize the risk of failure.
- Most of the overall risk of failure comes from taking on too much of the extract transformation-load (ETL) data staging design and development effort.
- In many cases, first-level data marts provide users with enough interesting data and ROI while the data mart teams focus on more difficult issues.



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Multisource Marts

- *Step 2*. Identify more complex multi-source marts
- Referred to as *consolidated data marts* because they typically cross business processes.
- It is prudent to focus on the first-level data marts as dimensional building blocks before tackling the task of consolidating.
- In some cases, the consolidated data mart is a *simple union of data sets* from the first-level data marts.
- Thus, it can often be easier to rebuild the consolidation *within* the data warehouse.



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Using the Matrix

- The columns of the matrix represent the common *dimensions* used across the enterprise.
- *Maybe create a comprehensive list of dimensions before filling in the matrix*?
- **X** indicates that the dimension column is related to the business process row.
- Looking across the rows captures the *dimensionality* of each data mart at a glance.
- Visual power: examine the columns that depict the *interaction* between the *data marts* and *common dimensions*.



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Analyse the Matrix

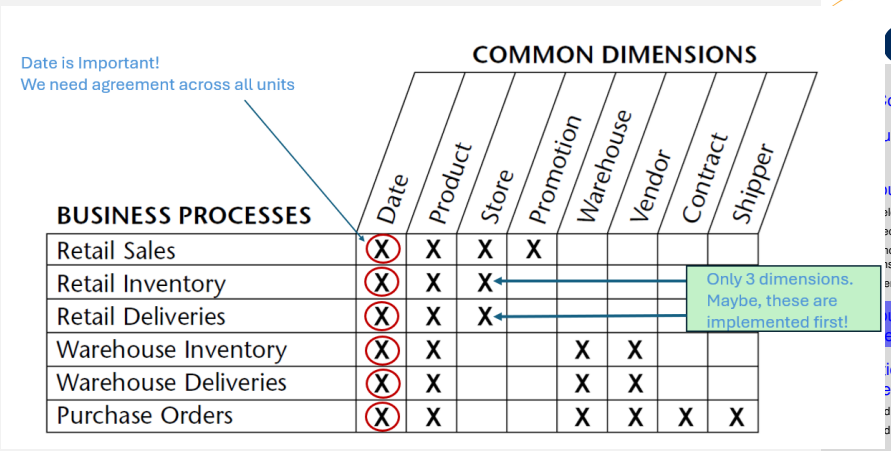


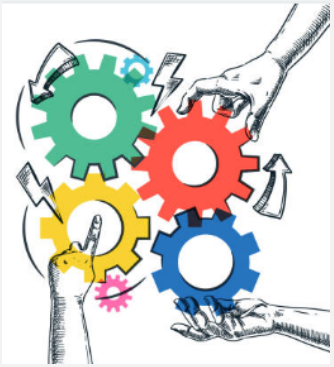
Figure 13: Decision Support

Matrix Benefits: planning and communication

- Defines the overall data architecture for the warehouse (by laying out rows & columns)
- *What is a high priority dimensions?* Participation in multiple data marts!
- Visually: communicate effectively within and across data mart teams.

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Integration and Interoperability



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A Framework for Integration and Interoperability

- It represents poor design to build separate data marts that ignore a *conforming* framework.
- Isolated, independent data marts deliver access to irreconcilable views of the organisation with reports that cannot be compared.
- Independent data marts become legacy implementations, blocking the development of a coherent warehouse environment.

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Conformed Dimensions

- Conformed dimensions are either identical or strict mathematical subsets of the most granular, detailed dimension.
- Conformed dimensions have:
  - consistent dimension keys,
  - consistent attribute column names,
  - consistent attribute definitions, and
  - consistent attribute values (which translates into consistent report labels and groupings).
- Non-conforming dimensions give invalid results when a monthly Dimension is joined with a weekly Fact Table.



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Conformed dimensions: Different Possibilities

- Ideally, there is only **one version** of each Dimension table.
- If not, conformed dimensions must have *identical meaning* with every possible fact table to which they are joined.
- The *date* dimension table connected to the *sales facts* is identical to the *date* dimension table connected to the *inventory facts*.
- In general, the date dimensions in both data marts will have:
  the same number of rows;
  same key values;
  same attribute labels;
  same attribute definitions;
  and same attribute values.



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Conforming and Grain

- Most conformed dimensions are defined naturally at the *most granular level possible*.
- The grain of the *customer* dimension will be the individual customer.
- The grain of the *product* dimension will be the lowest level at which products are tracked in the source systems.
- The grain of the *date* dimension will be the individual day.



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Different Grains

- Sometimes dimensions are needed at a rolled-up level of granularity.
- Why? Often the roll-up dimension is required because the fact table *represents aggregated facts* that are associated with *aggregated dimensions*.
- This would be the case if we had a weekly inventory snapshot *in addition* to our daily snapshot. *How can they use the same dimensions?*
- In other situations, the facts simply may be generated by another business process at a higher level of granularity.



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Conforming by Roll-up

- A sales process may capture data at the **atomic** *product level*, while forecasting uses data at the *brand level*.
- It is *not possible* to share a single product dimension across the two business processes because **granularity is different**.
- Product and brand dimensions can conform if the brand table were a *strict subset* of the atomic product table.
- Attributes common to both detailed and rolled-up dimensions, such as the *brand* and *category* descriptions, should be **labeled**, **defined**, and **valued** identically in both tables, as Figure 14. *Great example of Metadata for an example question!*



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- Roll-up dimensions conform to atomic dimension if they are a strict subset of that atomic dimension.

Product Dimensions
Product Key (PK)
Product Description
SKU Number (Natural Key)
Brand Description
Subcategory Description
Category Description
Department Description
Package Type Description
Package Size
Fat Content Description
Diet Type Description
Weight
Weight Units of Measure
Storage Type
Shelf Life Type
Shelf Width
Shelf Height
Shelf Depth
… and more

Conforms

Brand Dimension
Brand Key (PK)
Brand Description
Subcategory Description
Category Description
Department Description

Figure 14: Conforming Rollup Dimension Subsets

Data Warehouse and OLAP Technology



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Conforming Fact Tables

- There are three fundamental types of fact tables: **transaction**, **periodic snapshot**, and **accumulating snapshot**.
- These three fact table variations are not totally dissimilar because they share *conformed dimensions*.
- These are the keys to building separate fact tables that can be used together with common, consistent filters and labels.
- While the dimensions are shared, the usage and volume of the three fact tables differ.

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Transaction Fact Tables

- Most fundamental view of business's operations is at the *individual transaction level*.
- A row exists in the fact table for a given customer or product *only* if a transaction event occurred.
- On the other hand, a specific customer or product should be linked to *multiple* rows in the fact table as the customer or product is involved in multiple transactions.

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High Volume Detailed Data

- Transaction data fits easily into a dimensional framework.
- The lowest-level data is naturally dimensional, supporting analyses not possible on summarised data (detailed analysis).
- However, processing high volume transaction data can be slow!



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Periodic Snapshot Fact Tables

- **Periodic snapshots** present the *cumulative performance* at regular, predictable time intervals.
- Take a snapshot of the activity at the end of a day, week, or month, then another picture at the end of the next period, and so on.
- Each periodic snapshot is *stacked consecutively* into a **Periodic Snapshot Fact Table**.
- By the end of June there could be 6 monthly snapshots or 26 weekly snapshots.
- The is a traditional method for retrieval of a regular, predictable, trendable view of the key business performance metrics.



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Transaction and Periodic: same data!

- Data Warehouses will always have some level of **redundancy** with a lot of duplicated data.
- Periodic snapshots represent an *aggregation* of the transactional activity during that time period.
- Materialise the summary snapshot for performance reasons.
- Design of the snapshot and companion transaction tables takes place simultaneously.



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Accumulating Snapshot Fact Tables

- An **Accumulating Snapshot Fact Table** records a single row for an activity to be closely monitored, such as a problem product or a single customer's credit.
- They represent an *indeterminate* time span, covering the complete life of a transaction or *discrete* product (or customer).
- Accumulating snapshots have multiple date stamps, representing the predictable major events or phases that take place during the course of a lifetime.
- Includes an additional date column to record snapshot capture: facilitates time series analysis.



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Summary

CHARACTERISTIC	TRANSACTION GRAIN	PERIODIC SNAPSHOT GRAIN	ACCUMULATING SNAPSHOT GRAIN
Time period represented	Point in time	Regular, predictable intervals	Indeterminate time span, typically short-lived
Grain	One row per transaction event	One row per period	One row per life
Fact table loads	Insert	Insert	Insert and update
Fact row updates	Not revisited	Not revisited	Revisited whenever activity
Date dimension	Transaction date	End-of-period date	Multiple dates for standard milestones
Facts	Transaction activity	Performance for predefined time interval	Performance over finite lifetime

Figure 15: Fact Table Comparison



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